

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Any four (x1) from A. Xylem transports {water/minerals} {from roots / <u>up</u> the plant/ to the leaves}/ xylem allows transpiration (1) NOT nutrients B. Lack of water causes {leaf/ cells} to {lose turgor / become flaccid}/ cells become plasmolysed (1) C. Reduced surface area (1) D. {No/reduced} photosynthesis/ less water for photosynthesis (1) E. Less {products / named products} of photosynthesis (1) F. For respiration / growth (1)		2	2	4		
		(ii)		Phloem transport is bidirectional / up <u>and</u> down plant (1) (so) Symptoms appear in both roots and leaves (in potato leaf roll) (1)		2		2		
	(b)	(i)		A – sieve tube (element) NOT sieve plate B – companion cell (1) 1 mark for both	1			1		
		(ii)		A - lacks {nucleus / nucleolus } / {fewer/ no} mitochondria / ER / Golgi/ organelles/ less cytoplasm (1) ORA Must be clear which cell is referred to		1		1		
		(iii)		Plasmodesmata (1) {Loading / unloading/ transport/ transfer} of sucrose / Transport of {ATP / proteins / enzymes / macromolecules} (1)		2		2		
				Question 3 total	1	7	2	10	0	0